



THE HILLS GRAMMAR SCHOOL

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

2004

MATHEMATICS

EXTENSION 1

Teacher Responsible: Mrs P Singh

General Instructions:

- Reading time – 5 minutes
- Working time – 2 hours
- This paper contains 7 questions.
ALL questions to be attempted.
- ALL questions are of equal value.
- ALL necessary working should be shown
in every question in the booklets provided.
- Start each question in a new booklet.
- A table of standard integrals is supplied
at the back of this paper.
- Board approved calculators may be used.
- Hand up your paper in ONE bundle,
together with this question paper.
- ALL HSC course outcomes are being
assessed in this task. The Course Outcomes
are listed on the back of this sheet.

Students are advised that this is a Trial Examination only and cannot in any way guarantee the content or the format of the Higher School Certificate Examination.

Question One

Marks

- (a) Evaluate $\log_8 e$, correct to 2 decimal places. 1
- (b) Differentiate $y = \cos^3 x$. 2
- (c) If $\tan \theta = m$ and $\tan \phi = 3$, find the value of m if $\theta - \phi = \frac{\pi}{4}$. 3
- (d) $P(x) = x^3 - 3x^2 - 3x + 10$
- (i) Show that $x = 2$ is a root of $P(x)$. 1
- (ii) What is the product of the other 2 roots? 1
- (e) Evaluate $\int_0^{\frac{\pi}{3}} \tan^2 x \, dx$ (leave answer in exact form). 3
- (f) Write down the equation of the vertical asymptote of $y = \frac{3x}{2x-1}$. 1

$$\begin{aligned} 2x-1 &\neq 0 \\ 2x &\neq 1 \\ x &\neq \frac{1}{2} \\ x &= \frac{1}{2} \\ y &= \frac{3}{2 - \frac{1}{x}} \\ \text{when } x \rightarrow \infty, y &\rightarrow \frac{3}{2} \\ x \rightarrow -\infty, y &\rightarrow \frac{3}{2} \\ y &= \frac{3}{2} \text{ hor -} \end{aligned}$$

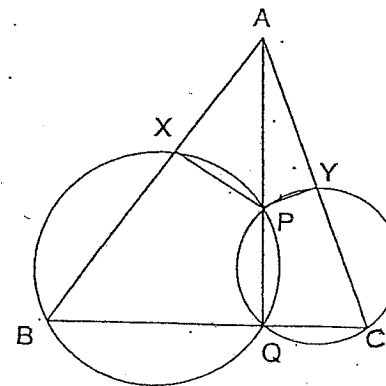
Question Two

- (a) Find the point which divides the line joining (4, 6) to (13, 5) externally in the ratio 4:1. 2
- (b) If the equation $3x^3 - 4x^2 + 2x + 1 = 0$ has roots α , β and λ , find
- (i) $2\alpha + 2\beta + 2\lambda$ 1
- (ii) the equation whose roots are 2α , 2β and 2λ . 2
- (c) Solve for x : $\frac{x^2 - 5x}{4 - x} \leq -3$. 3
- (d) Sand pouring from a pipe at a rate of $16\text{ m}^3 \text{ min}^{-1}$ forms a conical pile with height always equal to one quarter of the diameter of the base. (Volume of cone = $\frac{1}{3}\pi r^2 h$). How fast is the height of the pile rising at the instant when the pile is 4m high? 4

Question Three

- (a) Show that the function $f(x) = 5x - \sin 4x - 12$ is increasing for all values of x . 2
- (b) (i) Express $\sin x - \sqrt{3} \cos x$ in the form $A \sin(x - \alpha)$ with $A > 0$ and $0 < \alpha < \frac{\pi}{2}$. 2
- (ii) Find in exact form, the solution to $\sin x - \sqrt{3} \cos x = \sqrt{2}$ $0 < x < 2\pi$. 2
- (c) Prove, using the principle of Mathematical Induction, $\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1}$. 2

(d)



AXB, AYC, APQ and BQC are straight lines. Let $\angle ABC = x$ and $\angle ACB = y$. Copy this diagram in your answer booklet.

- (i) Prove $\angle XPY = x + y$; 2
- (ii) Prove AXPY is a cyclic quadrilateral. 2

Question Four

(a) (i) Show $\frac{d}{dx}(x \log_e x - x) = \log_e x$. 2

(ii) Hence, or otherwise, evaluate $\int_1^2 \log_e x \, dx$, in exact form. 2

(b) Evaluate $\int_0^1 \sqrt{1-x^2} \, dx$, by using $x = \sin \theta$, leaving answer in exact form. 4

(c) Given: $f(x) = x^2 + 1 \quad x \geq 0$

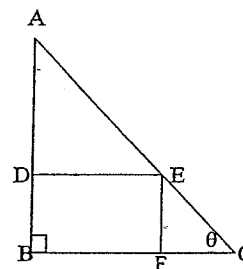
(i) Find $f^{-1}(x)$, stating the domain and range. 2

(ii) Sketch both $f(x)$ and $f^{-1}(x)$ on the same system of axes. 2

Question Five

(a) Prove that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$. 3

(b) Find the dimensions of the rectangle of maximum area which can be inscribed in a right angled triangle with sides 12, 16 and 20cm. One corner of the rectangle is to sit in the right angle of the triangle (see below). Let $EF = y$ and $DE = x$. 5



Copy this diagram into your answer booklet.

(c) (i) Evaluate $\int_1^3 \frac{dx}{x}$. 1

(ii) Use Simpson's Rule with 3 function values to approximate $\int_1^3 \frac{dx}{x}$. 2

(iii) Use your results to parts (i) and (ii) to obtain an approximation for e , to 3 decimal places. 1

Question Six

- (a) Given $f(x) = \frac{x}{x^2 - 1}$
- (i) Find the asymptotes and intercepts of $f(x)$. 3
 - (ii) Find the stationary points, if any, and determine their nature. 1
 - (iii) Find any points of inflection. 2
 - (iv) Sketch $f(x)$ showing the features from (i), (ii) and (iii). 3
- (b) (i) In how many ways can a committee of 3 women and 4 men be chosen from 8 women and 7 men? 1
- (ii) What is the number of ways a committee can be formed if woman A refuses to serve on the same committee as woman B. 2

Question Seven

- (a) A particle is oscillating in simple harmonic motion such that its displacement x metres from a given origin O satisfies the equation:

$$\frac{d^2x}{dt^2} = -4x \quad \text{where } t = \text{time in seconds.}$$

- (i) Show that $x = a \cos(2t + \beta)$ is a possible equation of motion for the particle when a and β are constants. 1
- (ii) The particle is observed at time $t = 0$ to have velocity of 2 m/s and a displacement from the origin of 4 m. Find the amplitude of oscillation. 3
- (iii) Determine the maximum velocity of the particle. 1

- (b) A cricket ball leaves the bowler's hand two metres above the ground with a velocity of 30 m/s at an angle of 5 degrees below the horizontal. The equations of motion for the ball are

$$\ddot{x} = 0 \quad \text{and} \quad \ddot{y} = -10.$$

Take the origin to be at the point where the ball leaves the bowler's hand, at $t = 0$.

- (i) Using calculus, prove that the coordinates of the ball at time t are given by

$$x = 30t \cos(5^\circ), \text{ and} \\ y = -30t \sin(5^\circ) - 5t^2. \quad 2$$

- (ii) Find the time at which the ball strikes the ground. 2
- (iii) Calculate the angle at which the ball strikes the ground. 2

- (c) Find the derivative of $y = \sqrt{x + \sqrt{x + \sqrt{x}}}$. 1

END OF PAPER

soln max

Q1 [12]

(a) $\frac{\ln e}{\ln 8} = 0.48 \checkmark$ (1) (PE1)

(b) $y = 3 \cos^2 x \cdot \sin x = -3 \cos^2 x \sin x$ (2) (PE5)

(c) $\tan \theta = m$ $\tan(\theta + \phi) = 1 \checkmark$
 $= \frac{\tan \theta + \tan \phi}{1 + \tan \theta \tan \phi} = 1$

$\frac{m-3}{1+3m} = 1 \checkmark$

$m-3 = 1+3m$

$m = -2 \checkmark$

(3) (HE1)

(d) (i) $P(x) = x^3 - 3(2)x^2 + 3(2)x + 10 = 0$ (1) (PE3)
 $x=0$ is a root

(ii) $2\alpha\beta = \frac{-10}{1} \checkmark$ (1)
 $\alpha\beta = -5$

(e) $\int_0^{\pi/3} \tan^2 x dx$
 $= \int_0^{\pi/3} [\sec^2 x - 1] dx$
 $= [\tan x - x]_0^{\pi/3}$
 $= (\sqrt{3} - \pi/3) - 0$
 $= \sqrt{3} - \pi/3$ (3) (HE4)

(f) $2x-1=0$
 $x = 1/2 \checkmark$ (1) (PE3)

Q2 (12 marks)

(a) $X_T = \frac{-4+52}{3}$
 $= 16 \checkmark$

$Y_T = \frac{-6+20}{3}$
 $= 4\frac{2}{3} \checkmark$ [2, PE3]

$\therefore$ pt (16, 4 $\frac{2}{3}$)

(b) $\alpha + \beta + \gamma = 7/3$

$\alpha\beta + \alpha\gamma + \beta\gamma = 7/3$

$\alpha\beta\gamma = -1/3$

$\therefore 2(\alpha + \beta + \gamma) = 14/3 \checkmark$ [1, PE3]

(ii) $x^3 - (2\alpha + 2\beta + 2\gamma)x^2 + (\alpha\beta + \alpha\gamma + \beta\gamma)x = x^3 - \frac{8}{3}x^2 + \frac{8}{3}x + \frac{8}{3} = 0$
 $3x^3 - 8x^2 + 8x + 8 = 0 \checkmark$ (2)

(c) $\frac{x^2-5x}{1-x} \leq -3$ ($x \neq 1$)

$(4-x)(x^2-5x) \leq -3(4-x)^2$

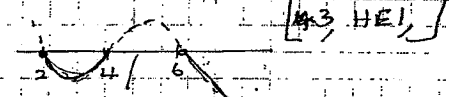
$(4-x)(x^2-5x) + 3(4-x)^2 \leq 0$

$(4-x)[x^2-5x+12-3x] \leq 0$

$(4-x)(x^2-8x+12) \leq 0$

$(4-x)(x-3)(x-6) \leq 0$

$\therefore 2 \leq x \leq 4$ and $x \geq 6$ (3)



(d) we need $\frac{dh}{dt}$

$h = \frac{1}{4}D$
 $= \frac{1}{4} \cdot 2r$
 $= \frac{1}{2}r$
 $\therefore r = 2h$ $\checkmark$

$V = \frac{1}{3}\pi(2h)^2 h$

$V = \frac{4}{3}\pi h^3 \checkmark$

$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$
 $= 4\pi h^2 \cdot \frac{dh}{dt} \checkmark$

$16 = 4\pi(4)^2 \cdot \frac{dh}{dt}$ [4, i]

$\therefore \frac{dh}{dt} = \frac{1}{4\pi} \text{ m}^3/\text{s}$

$\approx 0.080 \text{ m}^3/\text{s}$ (2dp)

Ques 2. (12 marks)

$$(a) f(x) = 5 - 4 \cos 4x \geq 5 - 4(1) \quad \checkmark \quad \text{since } -1 \leq \cos 4x \leq 1$$

$$f(x) \geq 1$$

$\therefore f(x)$ increases $\forall x$

[2, HE 1]

b) (1)

$$A \sin(x-x) = A \cos x \sin x - A \sin x \cos x = \sin x - \sqrt{3} \cos x$$

$$\Rightarrow A \cos x = 1$$

$$A \sin x = \sqrt{3}$$

$$\therefore \tan x = \sqrt{3}$$

$$x = \pi/3$$

$$\checkmark \quad A = \frac{\sqrt{1^2 + (\sqrt{3})^2}}{2}$$

$$A = 2$$

$$\therefore 2 \sin(x - \pi/3)$$

$$(i) 2 \sin(x - \pi/3) = \sqrt{2}$$

$$\sin(x - \pi/3) = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

$$\therefore (x - \pi/3) = \pi/4 \text{ or } 3\pi/4$$

$$\therefore x = \frac{7\pi}{12} \text{ or } \frac{13\pi}{12}$$

[4, HE 1, 7]

c) Prove $n=1$

$$\text{LHS: } \frac{1}{1+1} = \frac{1}{2} \quad \text{RHS: } \frac{1}{1+1} = \frac{1}{2}$$

$\therefore$ True for $n=1$

Assume true for $n=k$ + prove true for $(k+1)$

$$I = \frac{1}{R(R+1)} + \frac{1}{(R+1)(R+2)} = \frac{(R+1)}{(R+2)}$$

$$\text{LHS: } \frac{R}{(R+1)} + \frac{1}{(R+1)(R+2)}$$

$$= \frac{R(R+2) + 1}{(R+1)(R+2)}$$

$$= \frac{R^2 + 2R + 1}{(R+1)(R+2)}$$

$$= \frac{(R+1)^2}{(R+1)(R+2)}$$

$$= \frac{(R+1)}{(R+2)}$$

$$= \text{RHS}$$

Since $P(k)$ and $P(k+1)$ is true, $\therefore$ true for $n=1, 2, \dots$

[2, HE 2]

(3d)

$$(i) \angle XPO = 180^\circ - x \text{ and } \angle YPO = 180^\circ - y \quad \text{opp's of cyclic quad suppl.}$$

$$\therefore \angle XPY = 360^\circ - [180^\circ - x + 180^\circ - y] \quad (\text{rev})$$

$$= x + y$$

$$= \angle QED$$

$$(ii) \angle BAC = 180^\circ - (x+y) \quad \dots (\text{sum of } \Delta ABC)$$

$$\text{now } \angle XPY + \angle BAC = (x+y) + 180^\circ - (x+y)$$

$$\therefore \angle XPY = 180^\circ \quad (\text{opp's suppl.})$$

[4, PE 3]

Q4. (12 marks)

$$(a) (i) f(x) = x \log_e x - x$$

$$f'(x) = x \cdot \frac{1}{x} + \log_e x - 1$$

$$= 1 + \log_e x - 1$$

$$= \log_e x$$

$$= \angle QED$$

$$(ii) \int_1^2 \log_e x \, dx$$

$$= \int_1^2 x \log_e x - x \, dx$$

$$= [x \log_e x - x]^2$$

$$= [2 \log_e 2 - 2] - (-1)$$

$$= \log_e 4 - 1$$

[4, HE 3]

$$(b) \int_0^1 \sqrt{1-x^2} \, dx$$

$$x = \sin \theta \quad \checkmark$$

$$\therefore dx = \cos \theta \, d\theta$$

Terminals

$$x=0, \sin \theta = 0 \Rightarrow \theta = 0$$

$$x=1, \sin \theta = 1 \Rightarrow \theta = \pi/2$$

$$= \int_0^{\pi/2} \sqrt{1-\sin^2 \theta} \cos \theta \, d\theta$$

$$= \int_0^{\pi/2} \cos \theta \cos \theta \, d\theta$$

$$= \int_0^{\pi/2} \cos^2 \theta \, d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} (1 + \cos 2\theta) \, d\theta$$

$$= \frac{1}{2} [\theta + \frac{1}{2} \sin 2\theta]_0^{\pi/2}$$

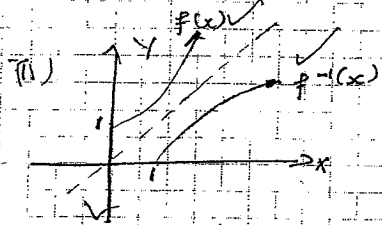
$$= \frac{1}{2} [\frac{\pi}{2}] = \frac{\pi}{4}$$

[4, HE 6]

4 Q (1) $p-16w = y^2 + 1$ ✓

$y^2 = x - 1$ ✓
 $y = \sqrt{x-1}$ ✓

Domain $x \geq 1$ ✓
 Range $y \geq 0$ ✓



[4, 11 E 4]

Q5 (2 marks)

Q) $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

LHS $\cos 3\theta = \cos(2\theta + \theta)$

$= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta$ ✓

$= (\cos^2\theta - \sin^2\theta)\cos\theta - 2[\sin^2\theta \cos\theta]$

$= \cos^3\theta - \cos\theta \sin^2\theta - 2\sin^2\theta \cos\theta$

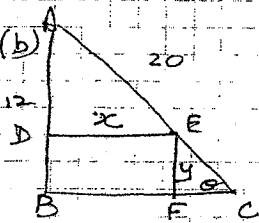
$= \cos^3\theta - 3\sin^2\theta \cos\theta$ ✓

$= \cos^3\theta - 3[1 - \cos^2\theta]$

$= \cos^3\theta - 3\cos\theta + 3\cos^3\theta$

$= 4\cos^3\theta - 3\cos\theta$

$= \text{RHS} \quad (3)$



$A = xy$

$\tan \theta = \frac{12}{16} = \frac{3}{4}$

$\tan \theta = \frac{y}{16-x}$ ✓

$\therefore y = \frac{3}{4}(16-x)$

$\therefore A = x \cdot \frac{3}{4}(16-x)$

$= 12x - \frac{3x^2}{4}$

$A' = 0$ for max area

$-\frac{3}{2}x = -12$ ✓

$\therefore x = 8 \text{ cm}$

$\therefore y = 6 \text{ cm} \quad \checkmark$

check for max

$A'' = -\frac{3}{2}$ ✓

$\therefore \cap < 0$ ✓

$\therefore$ max area

(PE 4)

Q5

(i) $\int_1^3 \frac{dx}{x}$

$= [\ln x]_1^3$ ✓

$= \ln 3$ ✓

(1)

iv)

x	1	$\frac{2}{3}$	$\frac{1}{3}$
y	1	$\frac{1}{2}$	$\frac{1}{3}$

$\int_1^3 \frac{dx}{x} = \frac{1}{3} [y_0 + 4y_1 + y_2]$

(2)

$= \frac{1}{3} [1 + 4 \times \frac{1}{2} + \frac{1}{3}]$

$= \frac{10}{9}$ ✓

ii) $\ln 3 = \frac{10}{9}$

$e^{\frac{10}{9}} = 3$

$e = 3^{9/10}$

(1)

$= 3^{0.9}$ ✓

(1E 1)

$= 2.688 \quad (3 \text{ dp})$

$$f(x) = \frac{x}{x^2-1}$$

(i) vertical asymptote $f(x) = \pm \infty$
 $x \rightarrow +1$ $f(x) = \pm \infty$
 $x \rightarrow -1$
 $\therefore x = \pm 1$ ✓

Horizontal

$$\lim_{x \rightarrow \pm \infty} \frac{x}{x^2-1} = 0$$

$$\therefore y = 0$$

Intercept
 $(0,0)$ ✓

(2)

(HE4)

(ii) $f'(x) = \frac{(x^2-1) \cdot 1 - x \cdot 2x}{(x^2-1)^2}$

$$= \frac{-x^2-1}{(x^2-1)^2}$$

$$f'(x) = 0$$

$$-x^2 = 1$$

$$x^2 = -1$$

No soln ✓

(1)

(iii) $f''(x) = \frac{2x(x^2-1)^2 - (-x^2-1) \cdot 2(x^2-1) \cdot 2x}{(x^2-1)^4}$

$$= \frac{(x^2-1)(-2x^3+2x+4x^3+4x)}{(x^2-1)^4}$$

$$= \frac{2x(x^2+3)}{(x^2-1)^3}$$

(2)

As inflection

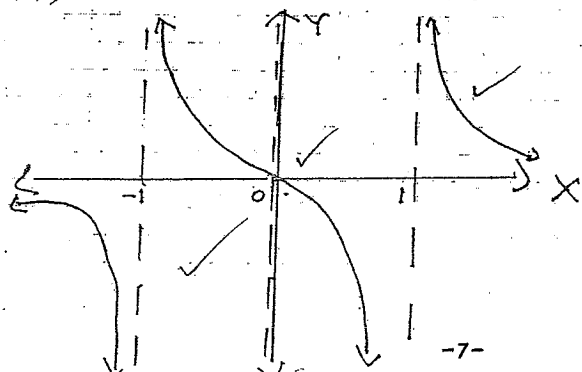
$$f''(x) = 0$$

$$\Rightarrow x = 0$$

$(0,0)$ ✓

$$x^2 = -3$$

(no soln)



(3)

(HE4)

Q6

(b) (i) 8P_3 (women) $\times$ 8P_4 men
 $= 1960$ ways. ✓

(1)

(ii) For both women A+B 6C_1 ways

$$\therefore {}^6C_1 \times {}^7P_4 = 210$$

✓

without both serving.

$$1960 - 210$$

$$= 1750 \text{ ways}$$

(2)

(PE3)

Q7

(a) (i) $y = \left[x + \left(x + (x^2)^{\frac{1}{2}} \right)^{\frac{1}{2}} \right]^{\frac{1}{2}}$

$$y' = \frac{1}{2} \left[x + \left(x + (x^2)^{\frac{1}{2}} \right)^{\frac{1}{2}} \right]^{-\frac{1}{2}} \cdot \left[1 + \frac{1}{2} \left(x + (x^2)^{\frac{1}{2}} \right)^{-\frac{1}{2}} \right] \cdot \left[1 + x^{-\frac{1}{2}} \right]$$

(HE5)

(b) (i) $x = a \cos \theta (2t + \beta)$
 $\dot{x} = -2a \sin \theta (2t + \beta)$
 $\ddot{x} = -4a \cos \theta (2t + \beta)$
 $= -4x$

[prove $\ddot{x} = -4x$]
 (1)

(ii) $t=0, x=2m/s, x=4m, a?$

$$x = a \cos \theta \dots (1)$$

$$2 = -2a \sin \theta$$

$$1 = -a \sin \theta \dots (2)$$

✓

(3)

$$(1)^2 + (2)^2$$

$$4 + 1 = a^2 \cos^2 \theta + a^2 \sin^2 \theta$$

$$17 = a^2$$

$$\therefore a = \sqrt{17}$$

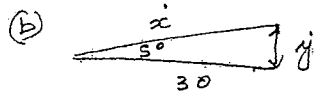
$$[\sin^2 \theta + \cos^2 \theta = 1]$$

(iii) $v = -2\sqrt{17} \sin \theta (2t + \beta)$

$$\therefore \max v = 2\sqrt{17} \text{ ms}^{-1}$$

(1)

(HE3)



when $t = 0$

$$\dot{x} = 30 \cos(5^\circ), \quad \dot{y} = -30 \sin 5^\circ \dots \textcircled{1}$$

(i) $\ddot{x} = 0$
 $\therefore \dot{x} = C_1$ (constant)

$$\therefore \dot{x} = 30 \cos 5^\circ \dots \textcircled{2} \text{ from } \textcircled{1}$$

$$x = \int 30 \cos 5^\circ dt \quad \checkmark$$

$$x = 30t \cos 5^\circ + C_2$$

when $t = 0, x = 0, \therefore C_2 = 0$

$$\therefore \boxed{x = 30t \cos(5^\circ)}$$

$$\ddot{y} = -10$$

$$\therefore \dot{y} = \int -10 dt$$

$$= -10t + D_1 \dots$$

when $t = 0, \dot{y} = -30 \cos(5^\circ) \dots \text{from } \textcircled{1}$

$$\therefore D_1 = -30 \sin(5^\circ)$$

$$\dot{y} = -10t - 30 \sin(5^\circ) \quad \checkmark \textcircled{3}$$

$$y = \int -10t - 30 \sin(5^\circ) dt$$

$$= -5t^2 - 30t \sin(5^\circ) + D_2 \quad \textcircled{2}$$

when $t = 0, y = 0 \therefore D_2 = 0$

$$\boxed{y = -30t \sin(5^\circ) - 5t^2} \dots \textcircled{4}$$

(ii) Ball strikes the ground when $y = -2$

Sub $y = -2$ in $\textcircled{4}$

$$-2 = -30t \sin 5^\circ - 5t^2$$

$$5t^2 + 30t \sin 5^\circ - 2 = 0$$

$$t = \frac{-30 \sin 5^\circ \pm \sqrt{(30 \sin 5^\circ)^2 - 4 \times 5 \times (-2)}}{2 \times 5}$$

$$= \frac{-30 \sin 5^\circ + \sqrt{900 \sin^2 5^\circ + 40}}{10} \quad \text{(ignore -ve)} \quad \tan \theta = \frac{\dot{y}}{\dot{x}} \quad \checkmark$$

$$= 0.4229 \text{ sec}$$

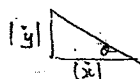
$$\approx 0.42 \text{ sec} \quad \checkmark$$

(2)

(iii) $t = 0.42 \text{ sec}$

$$\dot{x} = 30 \cos 5^\circ \text{ for}$$

$$\dot{y} = -4.229 - 30 \sin 5^\circ$$



$$\tan \theta = \frac{\dot{y}}{\dot{x}} \quad \checkmark$$

$$\theta = 12.9^\circ$$

$$\therefore \theta = 13^\circ \quad \checkmark$$